

Team Registration

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Project Idea 1

Project Topic Title:

Predicting High-Cost Medicare Patients Before They Become High-Cost

Dataset Description:

CMS Medicare Public Use Files (<https://data.cms.gov/>), a U.S. government Medicare claims data containing beneficiary demographics, chronic conditions, inpatient and outpatient utilization, and annual spending across millions of beneficiaries, available as an open download.

Project Idea Proposal:

1. Clear Health Question

Can we predict which Medicare beneficiaries will fall in the top 5% of total annual spending in the following year, using only their prior claims history?

The top 5% of Medicare beneficiaries account for approximately 50% of total Medicare expenditures. Health systems and ACOs operating under value-based care arrangements bear direct financial responsibility for these costs, yet they typically identify high-cost patients only after expenditures have been incurred leaving no window for proactive intervention.

2. Proposed Analytical Approach

Patient-level features will be constructed from Year 1 claims data including diagnosis codes, hospitalization frequency, chronic condition flags, comorbidity burden indices (Charlson/Elixhauser), and prior spending to train binary classification models predicting top 5% spending status in Year 2. A Logistic Regression model will serve as the interpretable baseline, followed by XGBoost as the primary predictive model. SHAP-based explainability will identify which patient characteristics most strongly drive predictions. Model performance will be evaluated using AUC-ROC, Precision at top K, and Recall, and benchmarked against published literature.

3. Health Relevance and Motivation

Early identification of high-risk patients enables targeted care management interventions such as nurse outreach, medication adherence programs, and preventive visits that can meaningfully reduce avoidable hospitalizations and generate measurable cost savings for health systems operating under risk-based contracts.

Project Idea 2

Project Topic Title:

Identifying Drivers of Healthcare Expenditure at the Household Level

Dataset Description:

Medical Expenditure Panel Survey (MEPS, <https://meps.ahrq.gov/mepsweb/>), a U.S. government survey providing individual and household-level data on healthcare utilization, expenditures, insurance coverage, and chronic conditions for approximately 35,000 individuals per year, available as an open download.

Project Idea Proposal:

1. Clear Health Question

What are the primary individual and household-level factors that drive high healthcare expenditure, and can we predict which individuals will incur high costs in a given year?

Understanding the determinants of healthcare spending is essential for insurers, employers, and policymakers designing cost-containment strategies. MEPS captures detailed longitudinal data on spending, utilization, and insurance type, making it well-suited for this analysis.

2. Proposed Analytical Approach

A healthcare cost prediction model will be built using individual-level features including insurance coverage type, chronic condition burden, demographic characteristics, and prior utilization. Logistic Regression and gradient boosting models (XGBoost or LightGBM) will be applied for cost classification, alongside a regression model to predict continuous expenditure levels. SHAP analysis will decompose the relative contribution of each feature to predicted costs. Results will be compared to published benchmarks from the health economics literature.

3. Health Relevance and Motivation

This project directly informs benefit design, care management targeting, and health equity initiatives providing actionable guidance to a health insurer or benefits manager seeking to reduce avoidable utilization among high-risk members.